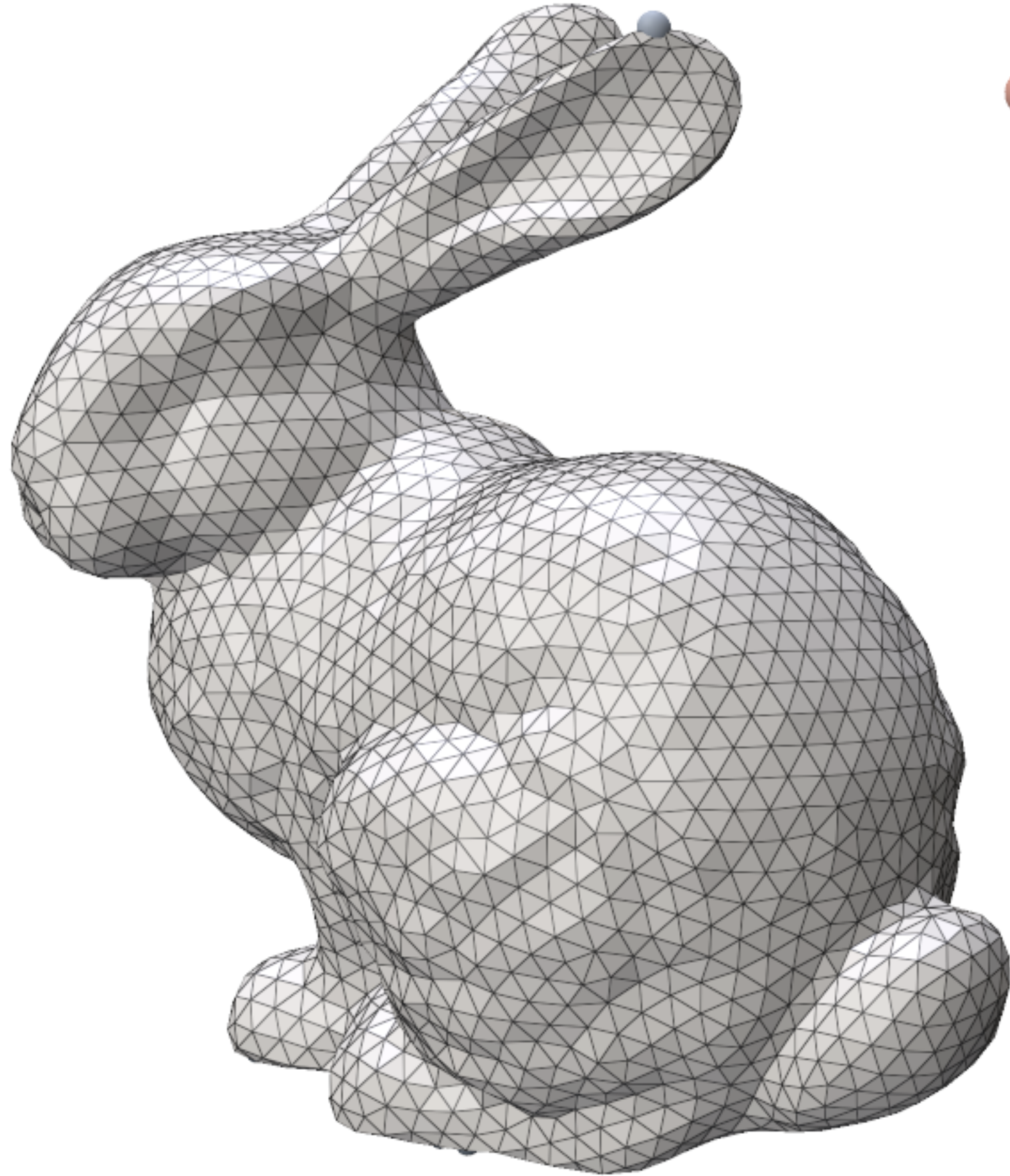
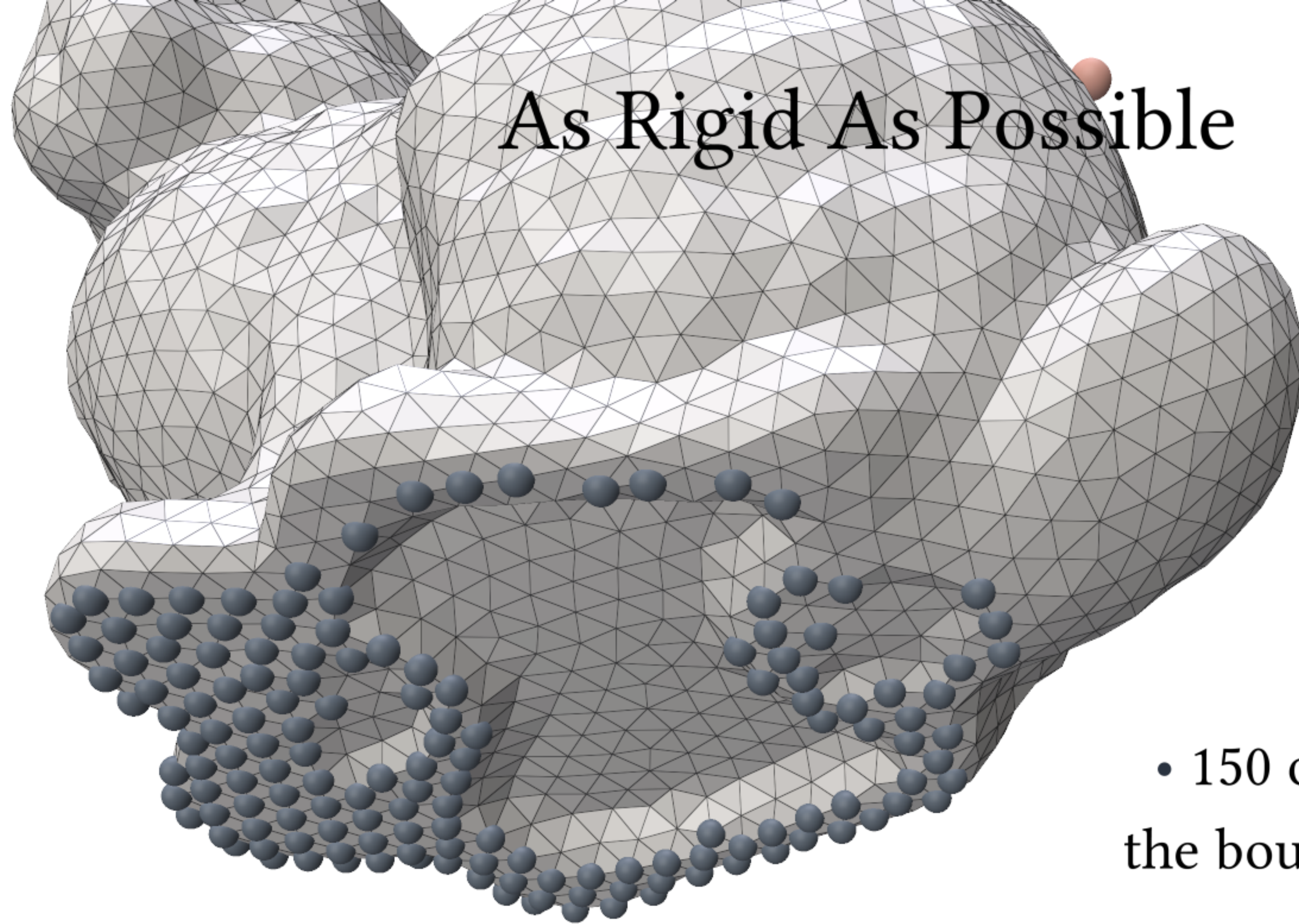


As Rigid As Possible



- the pinned base
- where you send it

Fix the base, then move one vertex.
The surface follows, as rigidly as it can.

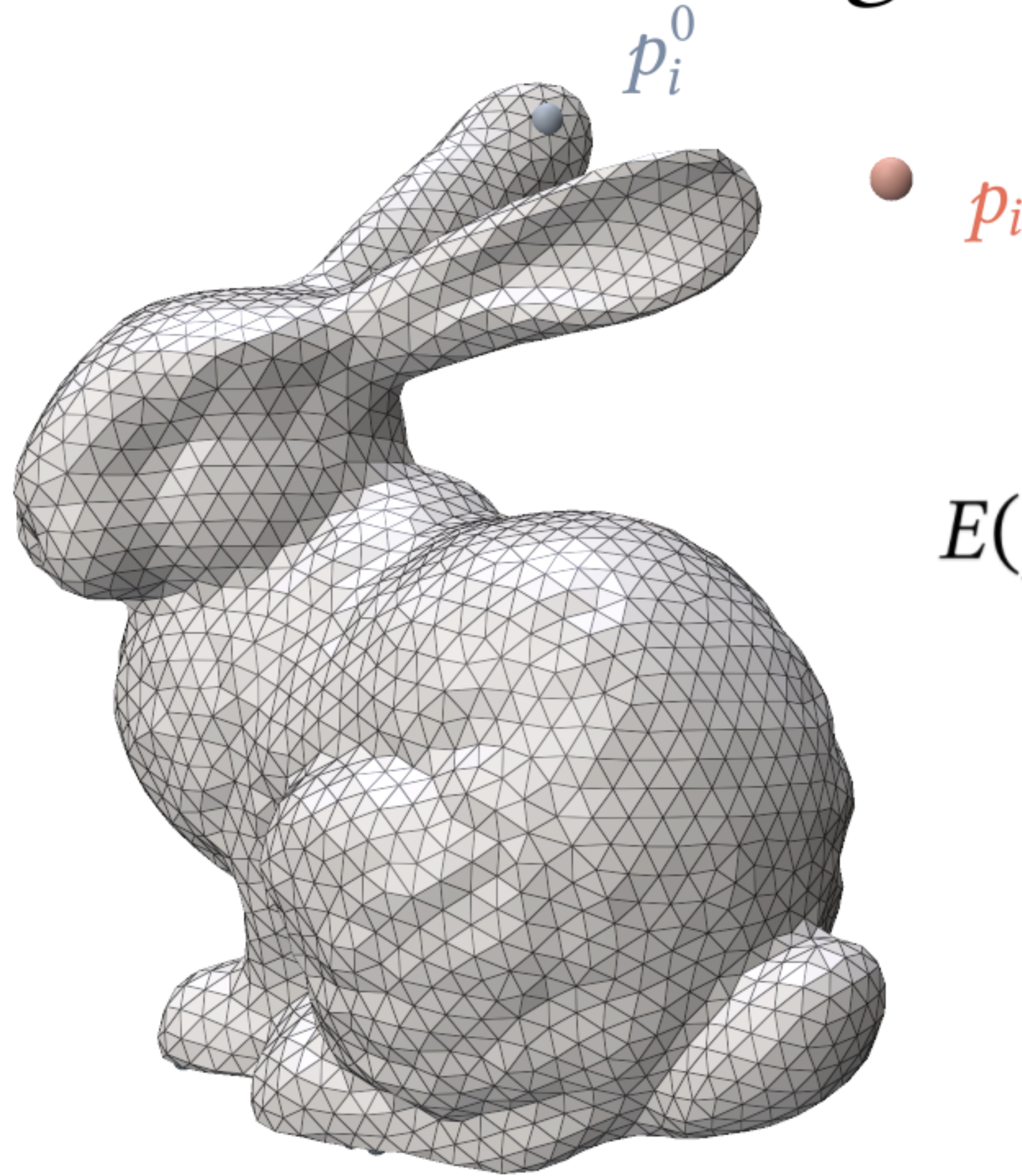


As Rigid As Possible

- 150 of 3000 vertices
the boundary condition

The base is pinned: these vertices never move.
Everything above them is solved for.

As Rigid As Possible

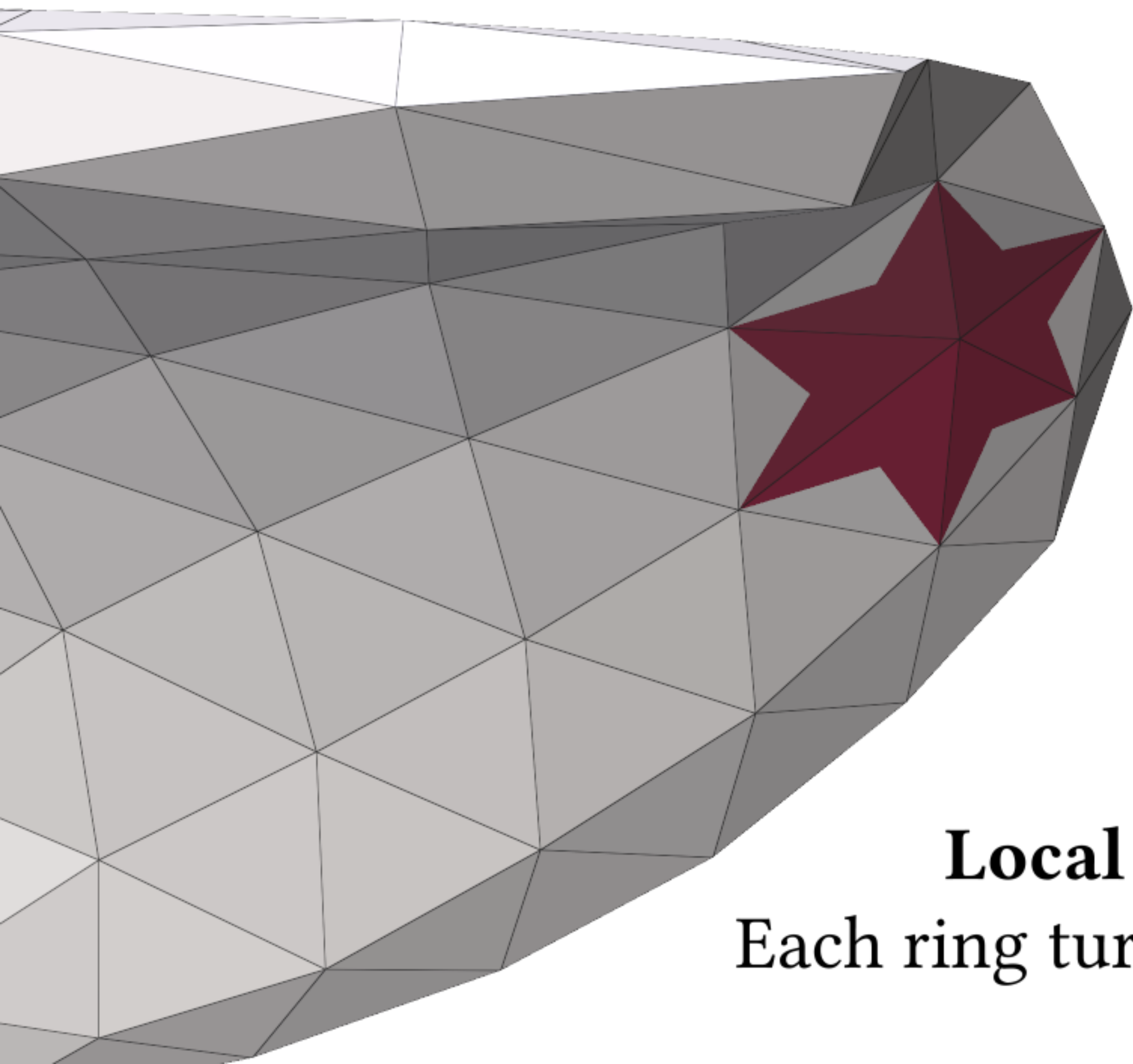


$$E(p) = \sum_i \sum_{j \in \mathcal{N}(i)} w_{ij} \|(p_i - p_j) - R_i (p_i^0 - p_j^0)\|^2$$

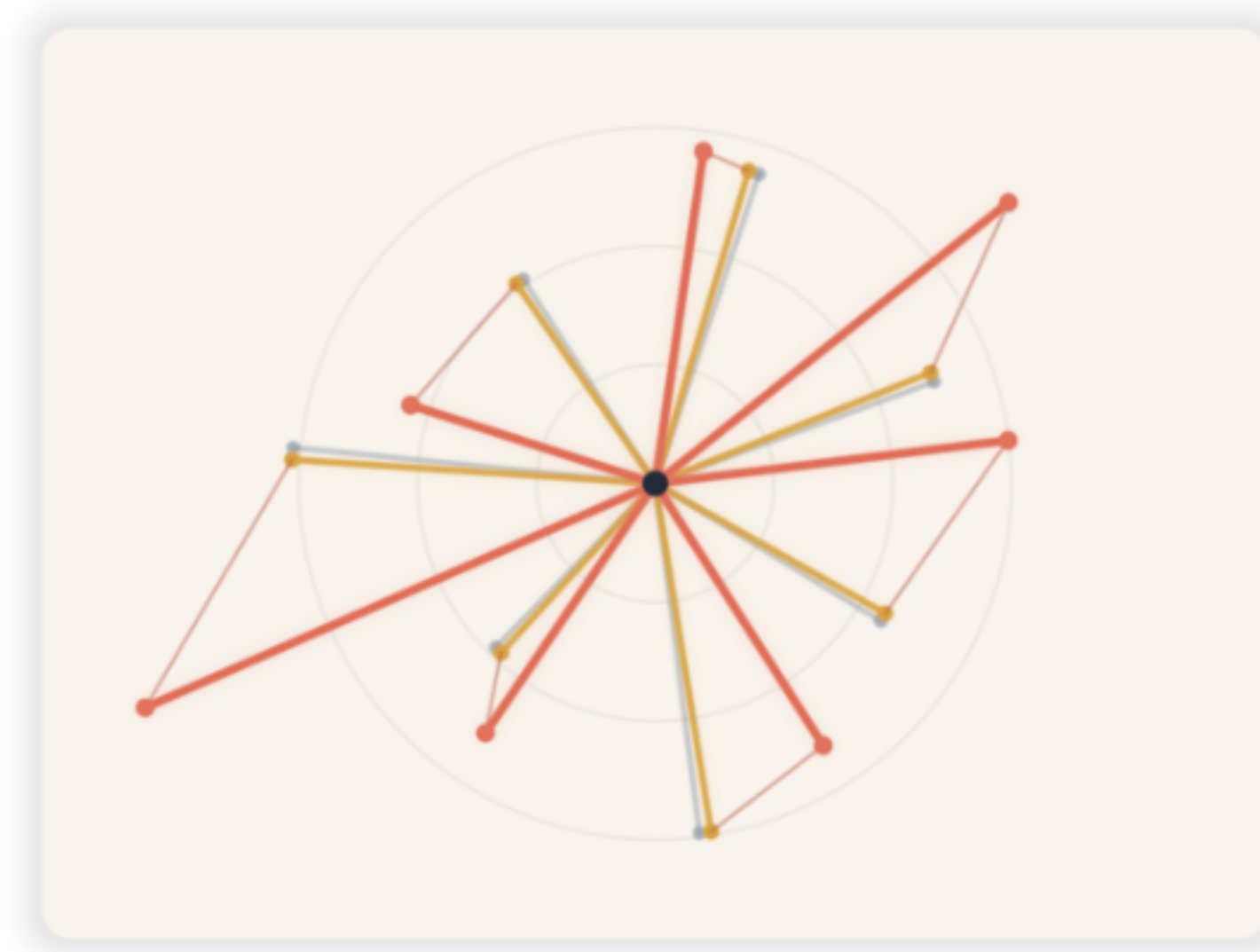
Every one-ring should move by a *rotation*.
The w_{ij} are cotangent weights.

As Rigid As Possible

$$S_i = \sum_j w_{ij} e_{ij}^0 e_{ij}^\top = U \Sigma V^\top \quad R_i = V U^\top$$



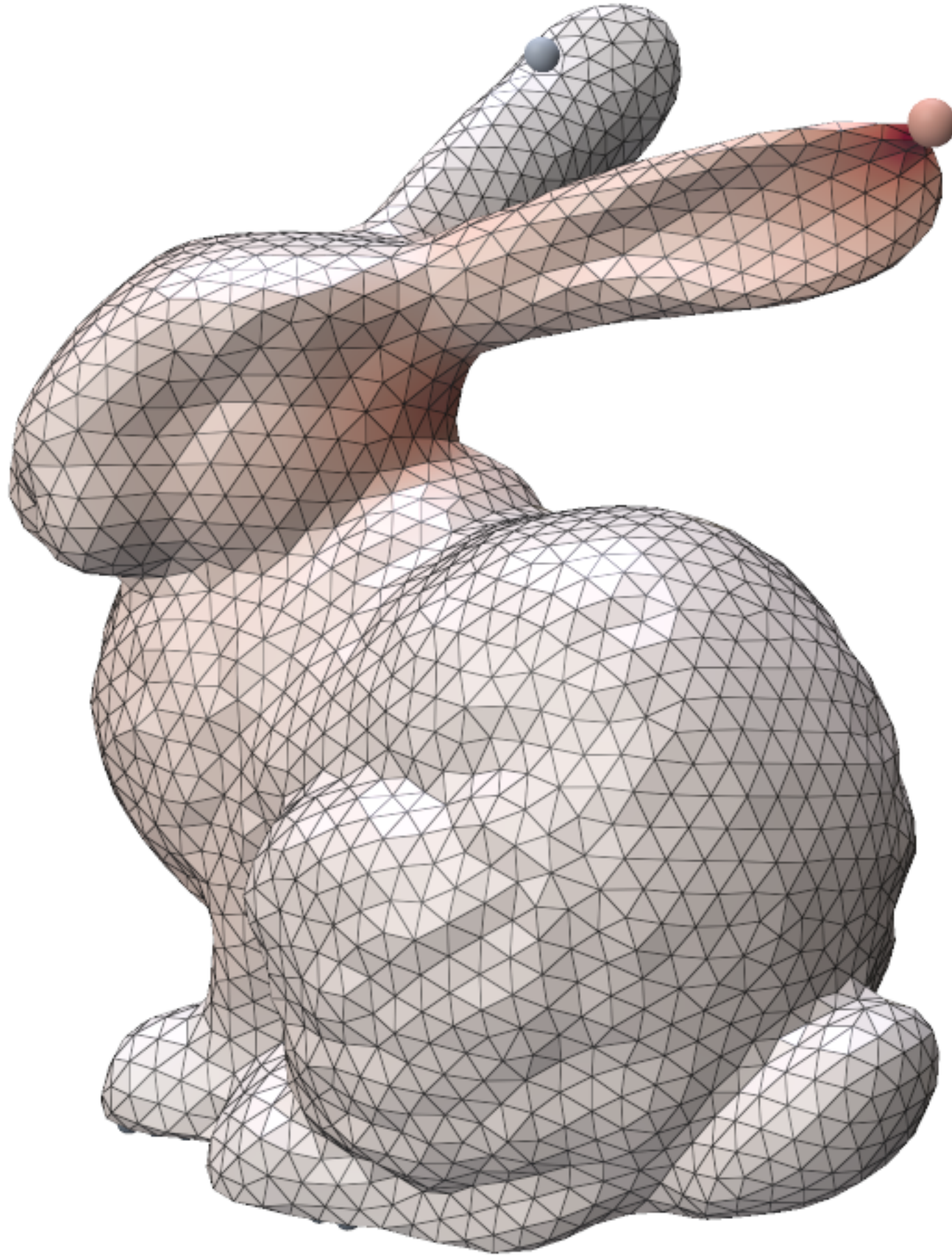
one ring, in 2D



e^0 rest e now Re^0 fitted

Local step. Freeze the positions.
Each ring turns by the rotation that fits it best.

As Rigid As Possible

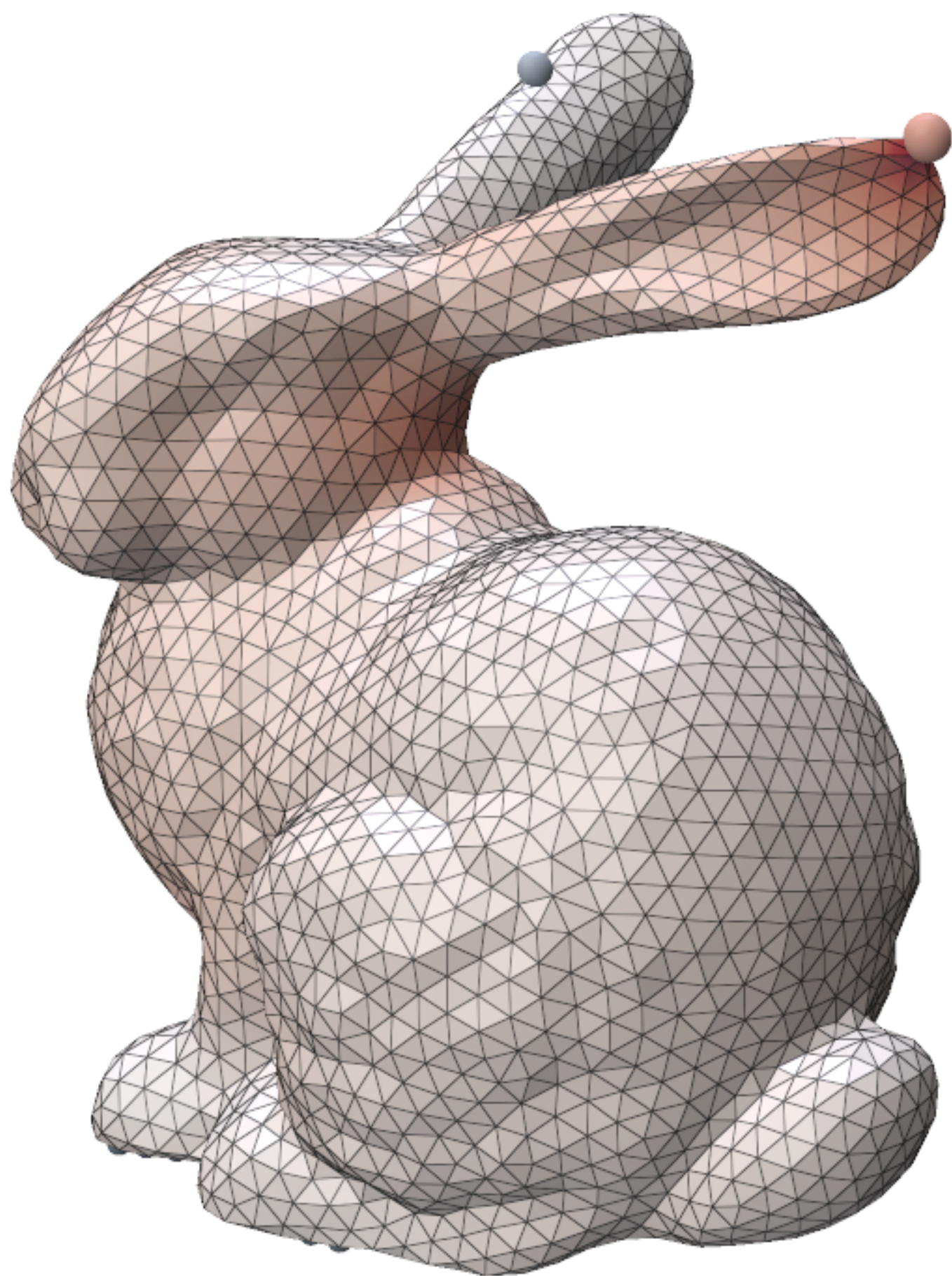


$$\sum_j w_{ij} (p_i - p_j) = \sum_j \frac{w_{ij}}{2} (R_i + R_j) (p_i^0 - p_j^0)$$

one Cholesky, at startup
a back-substitution per frame

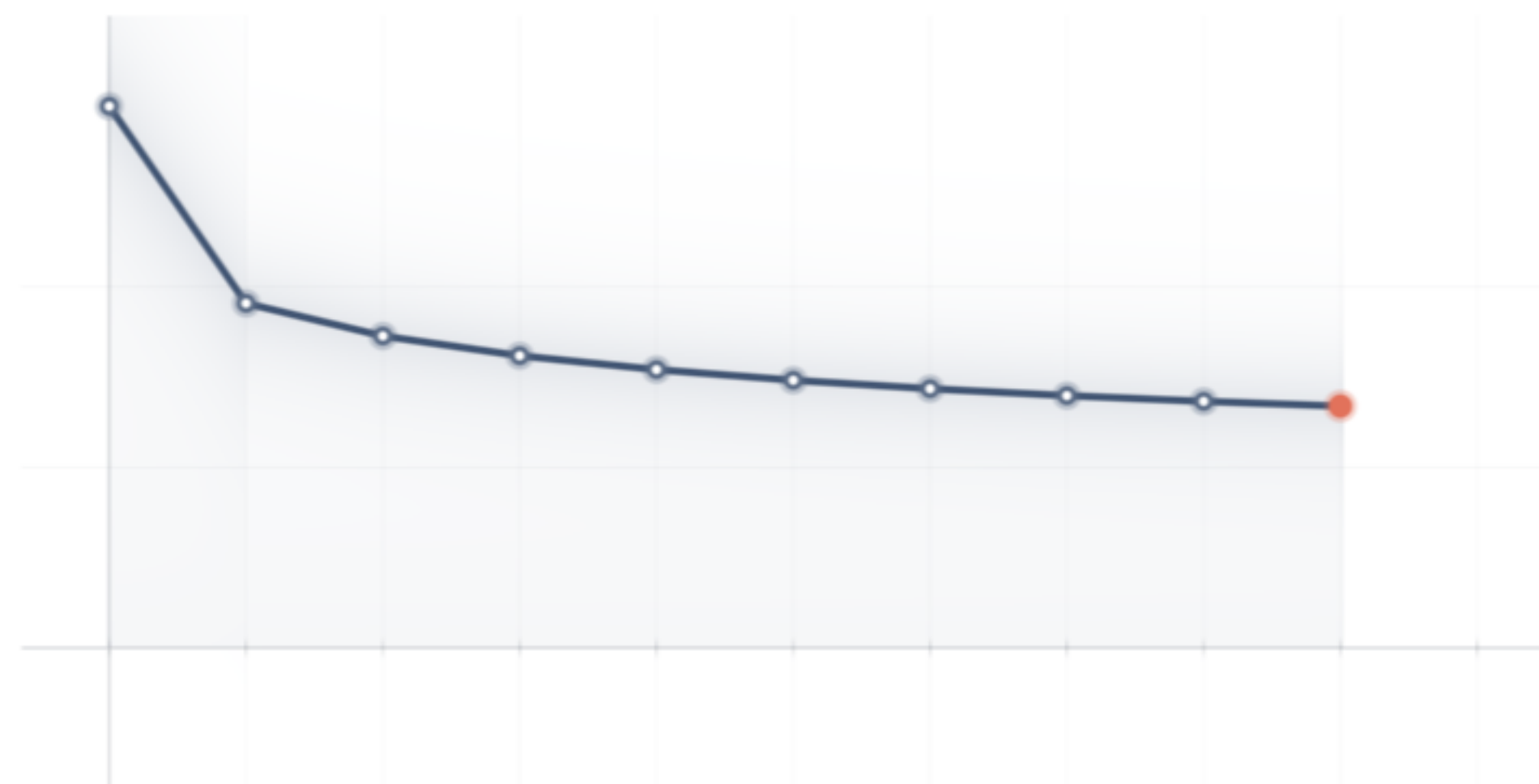
Global step. Freeze the rotations.
One sparse solve — factored once, reused every frame.

As Rigid As Possible



$$E(p^{(0)}) \geq E(p^{(1)}) \geq E(p^{(2)}) \geq \dots$$

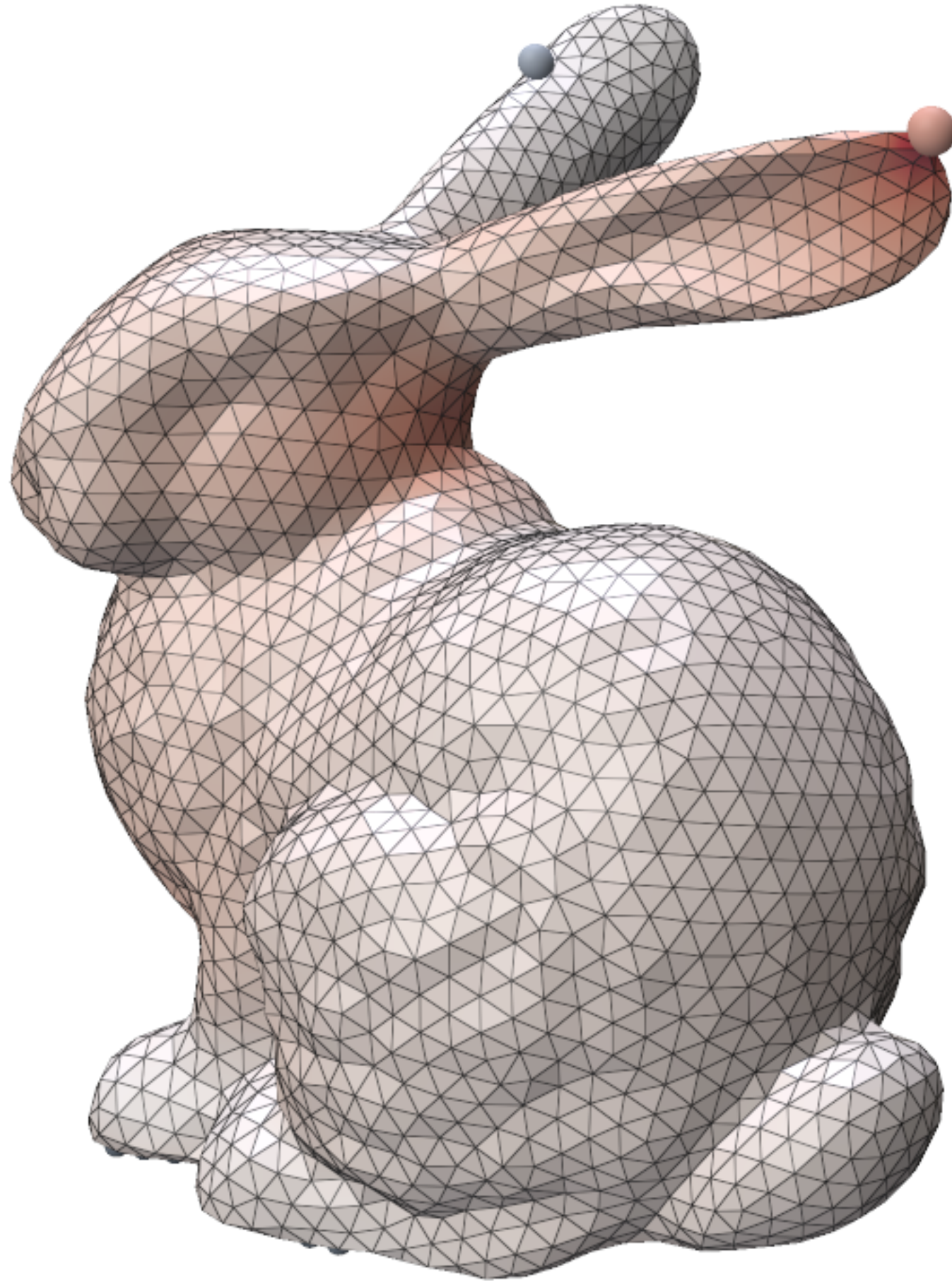
E/E_0 , after each sweep



sweeps

Neither step can raise E .
So alternating them walks it downhill.

As Rigid As Possible



arap/handle — drag the gizmo

arap/sweeps — how many, in the corner

Now grab it: the gizmo is already on the ear.
Every sweep is re-solved as you drag.